

PROFESSIONAL SUMMARY

MERN stack developer with experience building production-grade backend systems — architected a self-hosted PaaS with BullMQ-based CI/CD pipeline orchestration, an adaptive HLS video transcoding pipeline capable of handling highly concurrent queues and processing heavy raw video data reliably, and a published npm CLI with **1,076+ weekly organic downloads**. Strong foundation in Electronics & Communication Engineering with a keen interest in system design, real-time data pipelines, and secure web architectures.

TECHNICAL SKILLS

Languages:	C++, JavaScript (ES6+), Python
Frontend:	React.js, Redux Toolkit, Tailwind CSS, Three.js, GSAP, Framer Motion, Lenis
Backend:	Node.js, Express.js, REST APIs, JWT, OAuth, Passport.js, Bcrypt, RBAC, PM2
Databases & Caching:	MongoDB, Mongoose, Redis, BullMQ
Realtime & Video:	Socket.io, HLS, Fluent-FFmpeg, Adaptive Bitrate
Cloud & DevOps:	Cloudflare R2, Cloudinary, Vercel, GitHub Actions, CI/CD, Artillery, Autocannon
Security & Payments:	OAuth 2.0, Token Blacklisting, Razorpay
Developer Tools:	Commander.js, Archiver, Git, GitHub, Postman, npm

PROJECTS

NexForge — Self-Hosted PaaS Platform

Node.js, BullMQ, Redis, Socket.io, Express, React, GSAP + Lenis, GitHub OAuth, PM2

- Architected a self-hosted PaaS backend exposing build, deployment, and telemetry APIs; load-tested with **Autocannon** and **Artillery** under 50 concurrent connections, achieving **128+ req/s** sustained throughput, **366ms median** latency, and **100% success rate** across 1,000 requests with zero errors.
 - Engineered a CI/CD pipeline using **BullMQ + Redis** to orchestrate **50+ simultaneous build jobs**, with PM2 lifecycle management and symlink-based artifact swapping for **zero-downtime rollbacks** during deployments.
 - Published **nexforge-cli** on npm with dual-token Personal Access Token authentication and dynamic artifact pruning, reducing deployment upload sizes by **~90%** through intelligent compression.
 - Integrated **Socket.io + Recharts** for real-time build logs and hardware telemetry with **sub-100ms state synchronization**, enabling live build and queue monitoring with dashboard visibility.
- Pipeline:** CLI archive + PAT upload → Redis/BullMQ queue → worker extract & npm build → Cloudflare R2 storage → MongoDB route update → Express reverse proxy → Socket.io telemetry.

Live | GitHub | npm

Adaptive HLS Video Streaming Platform

React, GSAP + Lenis, Fluent-FFmpeg, BullMQ, Redis, Cloudflare R2, MongoDB

- Built an **adaptive-bitrate HLS transcoding engine** supporting 1080p/720p/480p with multi-audio tracks and soft subtitles using BullMQ + Fluent-FFmpeg for asynchronous background processing of heavy raw video data.
 - Designed a decoupled video pipeline using BullMQ workers and Cloudflare R2 for chunk storage, keeping CPU-intensive transcoding workloads off the main API request path for improved responsiveness.
 - Load-tested browsing and API workflows with **Autocannon** and **Artillery** using **650 virtual users** generating 1,950 HTTP requests at **50 req/s** sustained throughput; achieved **99.89% success rate** with 2.14s median response time under constrained free-tier compute.
 - Engineered FFmpeg worker with **ffprobe** media analysis and a **split-filter pipeline** that decodes HEVC 10-bit video once and simultaneously encodes 1080p/720p/480p H.264 streams, saving **50%+ CPU** over sequential encoding. Fine-tuned presets and threads for 512MB RAM servers, **reducing processing time of 25-min episodes by ~45% (4h → 2.25h)** with zero OOM crashes; generates MP4 fallbacks, thumbnail sprite sheets, and batches segments to Cloudflare R2.
- Pipeline:** Raw video upload → BullMQ worker queue → FFmpeg probe (audio/subs) + transcode (1080p/720p/480p) → R2 chunk storage → HLS master.m3u8 adaptive playback → Socket.io live progress streaming.

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Crochella — Multi-Vendor 3D E-Commerce

React 19, Three.js, Node.js, Socket.io, BullMQ, Razorpay, Redis

- Engineered a scalable multi-tenant marketplace with **3+ separate dashboards** (customer/admin/delivery); real-time Socket.io order tracking with Leaflet maps; Razorpay payment processing with automated PDF invoice generation.
 - Built an interactive **3D product catalog** using Three.js, GSAP ScrollTrigger, Lenis, and Framer Motion, achieving **60+ FPS** during interactive scrolling; Chrome DevTools profiling observed 126.7 FPS during testing.
 - Architected live delivery-tracking system with **sub-100ms geospatial state synchronization** using Socket.io and Leaflet maps; decoupled heavy workloads through BullMQ queues; integrated Razorpay payments and Cloudinary CDN.
 - Optimized marketplace performance through Redis caching, rate-limiting, and NoSQL injection prevention; load-tested with **Autocannon** and **Artillery** reaching **354+ req/s** peak throughput with **50.9ms median** latency, and rate-limiter deflecting 3,200+ brute-force requests with zero server crashes.
- Pipeline:** Cart checkout → Razorpay payment + webhook verification → order split into vendor sub-orders → BullMQ queue (receipts/payouts) → Socket.io vendor alerts → delayed escrow payout release.

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OPEN SOURCE

nexforge-cli

Published CLI for one-command application deployment using JWT-secured Personal Access Tokens and Archiver compression, achieving **1,076+ weekly organic downloads**.

EDUCATION

NIT Bhopal

B.Tech, Electronics & Comm. Engg. (2023–2027)
CGPA: 7.06/10
Relevant Coursework: DBMS, Operating Systems

Class XII (CBSE) — 77.0%

Class X (CBSE) — 92.0%

ACHIEVEMENTS & LEADERSHIP

- LeetCode:** Solved 500+ problems (287+ Medium, 82+ Hard) with **1,600+ Contest Rating**; maintained 238-day streak demonstrating consistent problem-solving practice.
- Robu Createo 3.0:** Secured 3rd Prize at national-level robotics and technical competition with 100+ participating engineering teams.
- Technical Executive, Evolve Club NIT Bhopal** (March 2024 – Present): Led project development teams, coordinated technical workshops, mentored junior members; led team to 3rd Prize at Robu Createo 3.0.